Software Engineering
Assignment-3
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following statements are true regarding the Agile software development model?

a. Only one increment is typically planned and developed at a time
b. Long-term detailed plans are usually avoided
c. Every iteration must add major new functionality
d. A new software release is generally produced at the end of each iteration
e. Each release may be delivered to the customer for regular use
f. Agile development requires completion of the entire system before deployment

Correct Answer: a. Only one increment is typically planned and developed at a time.
b. Long-term detailed plans are usually avoided.
d. A new software release is generally produced at the end of each iteration.
e. Each release may be delivered to the customer for regular use.

Detailed Solution:

Agile focuses on planning and developing one increment at a time, avoids long-term detailed plans, produces a release at the end of each iteration, and encourages delivering releases to customers for regular feedback. It does not require every iteration to add major functionality (c), nor does it require the entire system to be completed before deployment (f)—that is the opposite of Agile.

QUESTION 2:

A software company has recently developed an online shopping application for a retail store. The company has now been assigned a project to develop a similar application for a supermarket chain, with different billing rules and additional loyalty features. Which of the following software life cycle models would be most appropriate for this project?

a. Classical waterfall model
b. Prototyping model
c. Scrum
d. V model
e. Spiral model

Correct Answer: c. Scrum

Detailed Solution:

Scrum is most appropriate because the team already has a working baseline (the retail app) and can iteratively add/modify billing rules and loyalty features in short sprints, adapting to evolving requirements with frequent feedback. Waterfall, V-model, and pure prototyping are less flexible, and while Spiral is risk-driven, it is heavier and less practical than Scrum for this commercial adaptation project.

QUESTION 3:

Agile software development emphasizes which of the following principles?

a. Deliver working software quickly in small increments with adequate quality
b. Complete all documentation before starting implementation
c. Frequent customer feedback and continuous improvement
d. Deliver only fully completed systems at the end of the project
e. Respond effectively to changing requirements during development
f. Focus mainly on producing non-executable project documents rather than working software

Correct Answer: a. Deliver working software quickly in small increments with adequate quality
c. Frequent customer feedback and continuous improvement
e. Respond effectively to changing requirements during development

Detailed Solution:

Frequent changes, poor documentation, repeated patches, and lack of refactoring gradually degrade a software system's structure, making it harder to understand and modify over time—this is known as software entropy. The statement that continuous maintenance always improves structure is false, because without deliberate effort, maintenance often introduces new flaws and inconsistencies.

QUESTION 4:

Which of the following questions are typically discussed during a daily Scrum meeting?

a. What work did you complete since the last meeting?
b. What work will you complete before the next meeting?
c. What impediments are affecting your progress?
d. Who should be removed from the Scrum team?
e. What percentage of the project budget has been consumed?
f. What tasks are currently in progress?

Correct Answer: a. What work did you complete since the last meeting?
b. What work will you complete before the next meeting?
c. What impediments are affecting your progress?
f. What tasks are currently in progress?

Detailed Solution:

The Daily Scrum (stand-up) focuses on synchronizing the team by discussing work completed since the last meeting, work planned before the next meeting, impediments blocking progress, and tasks currently in progress. It is not a forum for team member evaluation (d) or budget tracking (e)—those are management concerns outside the daily stand-up.

QUESTION 5:

Which of the following statements are true regarding Scrum?

a. Scrum promotes iterative and incremental software development
b. A Scrum team organizes and manages its own work
c. Scrum requires all requirements to be frozen before development begins
d. Sprints are usually time-boxed iterations
e. The Scrum Master is responsible for writing all program code

Correct Answer: a. Scrum promotes iterative and incremental software development
b. A Scrum team organizes and manages its own work
d. Sprints are usually time-boxed iterations

Detailed Solution:

Scrum promotes iterative, incremental development with self-organizing teams and time-boxed sprints. It does not freeze requirements upfront (c)—it embraces change—and the Scrum Master is a facilitator, not the sole coder (e); the Development Team writes the code.

QUESTION 6:

Which of the following statements are true regarding the sprint backlog in a Scrum project?

a. It contains the tasks selected for the current sprint
b. It is owned and managed by the development team
c. It contains all desired work items for the entire project
d. It is updated throughout the sprint as work progresses
e. It is prepared and controlled exclusively by the Scrum Master
f. It mainly consists of detailed task-level work items

Correct Answer: a. It contains the tasks selected for the current sprint
b. It is owned and managed by the development team
d. It is updated throughout the sprint as work progresses
f. It mainly consists of detailed task-level work items

Detailed Solution:

The Sprint Backlog contains tasks selected for the current sprint, is owned and managed by the Development Team, is updated daily as work progresses, and consists of detailed task-level work items. It does not contain all work items for the entire project (that's the Product Backlog), and it is not controlled exclusively by the Scrum Master—the team owns it.

QUESTION 7:

Which of the following statements are true regarding Extreme Programming (XP)?

a. XP emphasizes Test-Driven Development (TDD)
b. Test cases are often developed based on user stories
c. XP encourages implementing features quickly and obtaining customer feedback
d. Refactoring is discouraged once the code is working
e. Customer feedback may lead to modification of implemented features
f. XP requires completing all features before any customer review takes place

Correct Answer: a. XP emphasizes Test-Driven Development (TDD)
b. Test cases are often developed based on user stories
c. XP encourages implementing features quickly and obtaining customer feedback
e. Customer feedback may lead to modification of implemented features

Detailed Solution:

XP emphasizes TDD, derives tests from user stories, encourages rapid implementation with frequent customer feedback, and allows feedback to modify features. It encourages continuous refactoring (not discourages it), and customer reviews happen throughout, not after all features are complete.

QUESTION 8:

Which of the following practices are commonly associated with Extreme Programming (XP)?

a. Frequent release of small software increments
b. Continuous customer involvement and feedback
c. Pair programming as a development practice
d. Avoiding code refactoring after implementation
e. Writing automated tests before or along with code
f. Freezing requirements at the beginning of the project

Correct Answer: a. Frequent release of small software increments
b. Continuous customer involvement and feedback
c. Pair programming as a development practice
e. Writing automated tests before or along with code

Detailed Solution:

XP practices include frequent small releases, continuous customer involvement, pair programming, and writing automated tests before or alongside code (TDD). It encourages continuous refactoring (not avoiding it) and embraces changing requirements (not freezing them upfront).

QUESTION 9:

Which of the following statements are true regarding roles in a Scrum project?

a. The Product Owner represents customer interests
b. The Development Team is usually cross-functional
c. The Scrum Master helps remove impediments faced by the team
d. The Scrum Master is solely responsible for coding the product
e. The Development Team typically consists of a small number of members
f. The Product Owner mainly acts as an external auditor of the project

Correct Answer: a. The Product Owner represents customer interests
b. The Development Team is usually cross-functional
c. The Scrum Master helps remove impediments faced by the team
e. The Development Team typically consists of a small number of members

Detailed Solution:

The Product Owner represents customer interests, the Development Team is cross-functional and small (typically 3–9 members), and the Scrum Master removes impediments. The Scrum Master is not solely responsible for coding (the team codes), and the Product Owner is deeply involved, not an external auditor.

QUESTION 10:

Which of the following statements are true regarding the product backlog in Scrum?

a. It contains all desired work items for the project
b. The product backlog may contain both story-based and task-based work items
c. The Product Owner is responsible for prioritizing the backlog
d. The sprint backlog and product backlog are always identical
e. The product backlog is updated as project needs evolve
f. Only completed work items are stored in the product backlog

Correct Answer: a. It contains all desired work items for the project
b. The product backlog may contain both story-based and task-based work items
c. The Product Owner is responsible for prioritizing the backlog
e. The product backlog is updated as project needs evolve

Detailed Solution:

The Product Backlog contains all desired work items (features, bugs, technical tasks), may include stories and tasks, is prioritized by the Product Owner, and is continuously updated as needs evolve. It is not identical to the Sprint Backlog (which is a subset for the current sprint), and it stores incomplete work—completed items are removed or marked done.

******THE END******